

Project Development Phase Model Performance Test

Date	03 October 2026
Team ID	SWTID-2026-5758
Project Name	Auto Ticket Classification Flow
Maximum Marks	4 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot																																																																																				
1.	Model Summary	Text classification model that reads a new ticket's short description and description, and predicts the Ticket Category (7 classes: Network / VPN, Access Management, Hardware / Laptop, Software / Application, Email / Collaboration, Password / Account, Security / Phishing). The prediction is used to route the ticket to the correct assignment group. Model: TF-IDF features (1–2 word n-grams) + Logistic Regression Data: 1,820 sample tickets, split 1,274 train / 273 validation / 273 test (70/15/15, stratified). Note: If the confidence score is 0.70 or higher, the ticket is classified and routed automatically. Below that, it goes to the Manual Triage queue for review.	Auto Ticket Classification - model pipeline <pre>graph TD A[New ticket (short description + description)] --> B[Text features TF-IDF (1-2 word n-grams)] B --> C[Classifier Logistic Regression, 7 classes] C --> D[Predicted class + confidence score] D --> E{confidence >= 0.70 ?} E -- Yes --> F[Auto-classify and route to assignment group] E -- No --> G[Manual Triage queue (needs review)]</pre>																																																																																				
2.	Accuracy	Training Accuracy: 100.00% Validation Accuracy: 99.63% Test Accuracy: 100.00% Precision / Recall / F1 (weighted, test set): 1.000 / 1.000 / 1.000 Note: These figures come from a prototype trained on a sample ticket dataset created for this project. The sample tickets are short and clearly worded, so scores are very high. Re-run on real ticket history before reporting final results.	Model evaluation <table><tr><td>Training accuracy</td><td>100.00% (1274 tickets)</td></tr><tr><td>Validation accuracy</td><td>99.63% (273 tickets)</td></tr><tr><td>Test accuracy</td><td>100.00% (273 tickets)</td></tr><tr><td>Precision (weighted)</td><td>1.000</td></tr><tr><td>Recall (weighted)</td><td>1.000</td></tr><tr><td>F1 score (weighted)</td><td>1.000</td></tr></table> Confusion matrix (test set) <table><tr><td></td><th>Access</th><th>Email</th><th>Hardware</th><th>Network</th><th>Password</th><th>Security</th><th>Software</th></tr><tr><th>Access</th><td>39</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><th>Email</th><td>0</td><td>39</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><th>Hardware</th><td>0</td><td>0</td><td>39</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><th>Network</th><td>0</td><td>0</td><td>0</td><td>39</td><td>0</td><td>0</td><td>0</td></tr><tr><th>Password</th><td>0</td><td>0</td><td>0</td><td>0</td><td>39</td><td>0</td><td>0</td></tr><tr><th>Security</th><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>39</td><td>0</td></tr><tr><th>Software</th><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>39</td></tr><tr><td></td><th>Access</th><th>Email</th><th>Hardware</th><th>Network</th><th>Password</th><th>Security</th><th>Software</th></tr></table> Predicted	Training accuracy	100.00% (1274 tickets)	Validation accuracy	99.63% (273 tickets)	Test accuracy	100.00% (273 tickets)	Precision (weighted)	1.000	Recall (weighted)	1.000	F1 score (weighted)	1.000		Access	Email	Hardware	Network	Password	Security	Software	Access	39	0	0	0	0	0	0	Email	0	39	0	0	0	0	0	Hardware	0	0	39	0	0	0	0	Network	0	0	0	39	0	0	0	Password	0	0	0	0	39	0	0	Security	0	0	0	0	0	39	0	Software	0	0	0	0	0	0	39		Access	Email	Hardware	Network	Password	Security	Software
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3.	Confidence Score (Only Yolo Projects)	Class Detected: Category predicted for the ticket, for example Network / VPN, or Access Management. Confidence Score: Probability the model gives to the predicted class. For example, 0.98 means the model is 98% sure the ticket is Network / VPN. Threshold: 0.70. On the test set, 99.3% of tickets scored at or above it (accuracy 100.0%) and 0.7% went to Manual Triage. Mean confidence: 0.93. Note: YOLO applies to object-detection projects only. This is a text classification flow, so the class probability is used as the confidence score. Vague tickets such as “it is not working” score low and go to Manual Triage.	<table border="1"> <caption>Predicted class and confidence score</caption> <thead> <tr> <th>Ticket Description</th> <th>Predicted Class</th> <th>Confidence Score</th> <th>Routing</th> </tr> </thead> <tbody> <tr> <td>cannot connect to vpn from home since this morning</td> <td>Network / VPN</td> <td>0.98</td> <td>Auto-routed</td> </tr> <tr> <td>need access to the shared finance folder for my team</td> <td>Access Management</td> <td>0.96</td> <td>Auto-routed</td> </tr> <tr> <td>laptop screen is flickering after it fell down</td> <td>Hardware / Laptop</td> <td>0.96</td> <td>Auto-routed</td> </tr> <tr> <td>possible phishing link was clicked please check</td> <td>Security / Phishing</td> <td>0.97</td> <td>Auto-routed</td> </tr> <tr> <td>forgot my password please help</td> <td>Password / Account</td> <td>0.94</td> <td>Auto-routed</td> </tr> <tr> <td>it is not working please check</td> <td>Hardware / Laptop</td> <td>0.27</td> <td>Manual Triage</td> </tr> <tr> <td>email access problem</td> <td>Security / Phishing</td> <td>0.37</td> <td>Manual Triage</td> </tr> </tbody> </table>	Ticket Description	Predicted Class	Confidence Score	Routing	cannot connect to vpn from home since this morning	Network / VPN	0.98	Auto-routed	need access to the shared finance folder for my team	Access Management	0.96	Auto-routed	laptop screen is flickering after it fell down	Hardware / Laptop	0.96	Auto-routed	possible phishing link was clicked please check	Security / Phishing	0.97	Auto-routed	forgot my password please help	Password / Account	0.94	Auto-routed	it is not working please check	Hardware / Laptop	0.27	Manual Triage	email access problem	Security / Phishing	0.37	Manual Triage
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